Sustainable Venturing

Entrepreneurial Opportunity in the Transition to a Sustainable Economy

Thomas J. Dean

SUSTAINABLE VENTURING

Entrepreneurial Opportunity in the Transition to a Sustainable Economy

Thomas J. Dean

Department of Management,
Colorado State University

With Contributions by Jacob T. Castillo

PEARSON

Boston Columbus Indianapolis New York San Francisco Upper Saddle River
Amsterdam Cape Town Dubai London Madrid Milan Munich Paris Montréal Toronto
Delhi Mexico City São Paulo Sydney Hong Kong Seoul Singapore Taipei Tokyo

Editor in Chief: *Stephanie Wall*
Acquisitions Editor: *Sarah Parker McCabe*
Director of Editorial Services: *Ashley Santora*
Editorial Project Manager: *Claudia Fernandes*
Editorial Assistant: *Ashlee Bradbury*
Director of Marketing: *Maggie Moylan*
Senior Marketing Manager: *Erin Gardner*
Marketing Assistant: *Gianni Sandri*
Production Project Manager: *Meghan DeMaio*
Creative Art Director: *Jayne Conte*
Cover Designer: *Karen Salzbach*
Full-Service Project Management: *Munesh Kumar*
Composition: *iEnergizer Aptara®, Inc.*
Printer/Binder/Cover Printer: *Courier Companies*
Text Font: *10/12 Times Ten LT Std*

Credits and acknowledgments borrowed from other sources and reproduced, with permission, in this textbook appear on the appropriate page within text.

Library of Congress Cataloging-in-Publication Data

Dean, Thomas J.,

Sustainable venturing : entrepreneurial opportunity in the transition to a sustainable economyb/ Thomas J. Dean, Department of Management, Colorado State University; written with the assistance of Jacob T. Castillo.

pages cm

ISBN-13: 978-0-13-604489-5

ISBN-10: 0-13-604489-1

1. Social entrepreneurship. 2. Entrepreneurship—Environmental aspects. 3. Technological innovations–Environmental aspects. 4. Social responsibility of business. I. Castillo, Jacob T. II. Title.

HD60.D347 2014

658.1′1—dc23

2013000194

10 9 8 7 6 5 4 3 2 1

PEARSON

ISBN 10: 0-13-604489-1
ISBN 13: 978-0-13-604489-5

Dedication

To my mother, Marie Dean

"*For I dipt into the future far as human eye could see,*
Saw the vision of the world and all the wonder that would be."

—Lord Alfred Tennyson

ALFRED, LORD TENNYSON, 1842. Locksley Hall, verses 6065,
The Poetical Works of Alfred, Lord Tennyson, p. 111

BRIEF CONTENTS

CONTENTS

PREFACE

Sustainable venturing is about harnessing the innovative power of entrepreneurship to solve global, social, and environmental challenges. From innovative new business models to new technical solutions and changing economic systems, opportunities for sustainable venturing abound. Sustainable venturing varies from local small business solutions with an environmental flair to emerging high-growth ventures delivering technologies in alternative energy, clean water, and efficiency. Regardless of their focus or magnitude, sustainable ventures are transforming the nature of business, and they can do so while providing financial return to entrepreneurs, their investors, and their organizations.

The prospect of doing well for the environment and society while operating an economically sustainable venture is exciting to potential entrepreneurs and global citizens alike. Increasingly, it seems students want to accomplish both good for society and good for themselves. This book attempts to show how that might be accomplished, and how entrepreneurship can be used as a transformative mechanism in our economic system.

At its core, this book is about engaging economic incentive systems to drive ecologically and socially sustainable behaviors and outcomes. The underlying philosophy is that the capitalistic system is a powerful means to create value for society, and the entrepreneur is the engine of that value creation. This text views capitalism and entrepreneurship as the solutions to our social and environmental challenges, and sustainable entrepreneurs as active participants in the creation of incentives for more sustainable behaviors.

Sustainable Venturing begins Chapter 1 with an introduction to the meaning and arena of sustainable venturing. Chapters 2 and 3 provide students with a broader understanding of the workings of economic systems and how entrepreneurs can interact with those systems, both capturing and creating opportunities in the dynamics of changing economic institutions. These chapters embrace concepts from the fields of environmental and welfare economics to help explain our global, social, and environmental challenges as well as the strategies entrepreneurs can use to overcome these challenges and create economic opportunity.

Chapter 4 is devoted to the exciting and growing concept of social entrepreneurship, which prioritizes social contribution in the development of new businesses. Chapter 5 turns to the practice of environmental marketing and outlines opportunities for the marketing of products with environmental attributes. It emphasizes the unique ways, sometimes unexpected, that organizations have succeeded by making environmental attributes part of their product and promotional strategies.

Chapters 6 and 7 focus on some of the practical aspects of creating a sustainable venture, covering how to launch and finance ventures with positive social and environmental goals and contributions. Chapter 6 covers six elements of the successful sustainable venture including mission, values, planning, legal form, team, and metrics. Chapter 7 discusses the process of funding and the unique sources of financing available to the sustainable venture.

Sustainable Venturing is designed for use in courses in sustainable, environmental, and social entrepreneurship. It serves as the perfect content companion to courses that integrate case studies, speakers, or experiential exercises to engage students in learning about the opportunities for entrepreneurship in sustainability. The text may also serve broader entrepreneurship courses wherein the instructor wishes to dedicate some portion of the class to social and environmental entrepreneurship. It may also be utilized within environmental science, policy, or management curriculum as an introduction to the business side of sustainability and environmental issues. It is written in a manner that is applicable to both graduate and undergraduate student audiences.

Sustainable Venturing covers a host of concepts, topics, and tools relevant to identifying opportunities and implementing a business with positive social and environmental outcomes. It also delivers examples of business models that have contributed to sustainability across a broad array of sectors. The examples demonstrate the methods used to create and capture value in the process of sustainable venturing. Finally, the book's focus on resolving system failures through entrepreneurial action provides a unique perspective, one that will hopefully lead sustainable entrepreneurs to thoughtful approaches that allow the simultaneous achievement of economic, social, and environmental sustainability.

SUPPLEMENTS

At www.pearsonhighered.com/irc, the following supplements are available to adopting instructors for download. Registration is simple and gives you immediate access to new titles and new editions. If you ever need assistance, our dedicated technical support team is ready to help with the supplements that accompany this text. Visit http://247.pearsoned.com/ for answers to frequently asked questions and toll-free user support phone numbers.

- Instructor's Manual
- PowerPoint Slides

COMPANION WEBSITE

A useful companion website, www.pearsonhighered.com/entrepreneurship, offers free access to teaching resources for all books in the Prentice Hall Entrepreneurship Series including additional activities, links to latest research, sample entrepreneurship curriculum and syllabi, teaching tips, and web resource links.

COURSESMART eTEXTBOOKS

CourseSmart eTextbooks were developed for students looking to save on required or recommended textbooks. Students simply select their eText by title or author and purchase immediate access to the content for the duration of the course using any major credit card. With a CourseSmart eText, students can search for specific keywords or page numbers, take notes online, print out reading assignments that incorporate lecture notes, and bookmark important passages for later review. For more information or to purchase a CourseSmart eTextbook, visit www.coursesmart.com.

ACKNOWLEDGMENTS

As with any significant endeavor, many individuals contributed to this book. I would first like to acknowledge the support and encouragement of Pearson Prentice-Hall's Entrepreneurship Series Editors, Drs. Duane Ireland and Michael Morris, without whose support and encouragement this book would not have been completed. I'm particularly grateful to Jacob Castillo, who wrote the first version of Chapter 6 and contributed editorially and conceptually to many other parts of the manuscript.

I would also like to thank those who contributed to the development of the ideas inherent in this book. In particular, authors in the fields of environmental economics and institutional economics laid the foundation for my intellectual contributions on the linkages between environmental issues and entrepreneurship theory.

The creativity and drive of sustainable entrepreneurs in Colorado and beyond have also been a source of encouragement, motivation, and validation. As they have succeeded in their ventures, I have been reassured that the ideas presented in *Sustainable Venturing* work in practice. I would also like to acknowledge some of those who supported me in the early days of my efforts at the intersection of environmental issues and entrepreneurship: Dale Meyer, David Payne, Alison Peters, and Paul Jerde.

My inspiration comes from my mother, a social entrepreneur whose example of caring, hard work, and seemingly never-ending willingness to serve is unparalleled in my entire life's experience. I credit my equanimity to both my father and my brother, who exemplify the characteristics of generosity, compassion, and integrity. My wife's love is the foundation for my ability to keep moving on the projects I pursue. My sons' laughter reminds me every day to smile and enjoy, and their words and futures encourage me to make a difference in the world.

ABOUT THE AUTHOR

Thomas J. Dean is Professor of Entrepreneurship and Sustainable Enterprise in the Department of Management at Colorado State University. He earned his PhD from the University of Colorado and his MBA from Oklahoma State University. He also received a BS in Environmental Resource Management from The Pennsylvania State University. He currently teaches in Colorado State University's innovative Global, Social, and Sustainable Enterprise MBA program. His previous academic appointments include the University of Colorado, where he served as the Faculty Director of the Deming Center for Entrepreneurship, and the University of Tennessee.

Dr. Dean created the first course in sustainable venturing in 2001 and has since gone on to publish pioneering conceptual work at the intersection of entrepreneurship, environmental issues, and sustainability. He also founded the first dedicated business plan competition for student clean technology ventures and sits on the advisory boards for the School of Global Environmental Sustainability and the Clean Energy Supercluster at Colorado State University. He also engages with student and nonstudent ventures and has worked in an executive capacity at a clean-tech venture. He served as Chair of the Entrepreneurship Division of the Academy of Management and founded one of the first business-school environmental management programs in the United States.

Dr. Dean has published articles on entrepreneurship and environmental issues in journals such as the *Academy of Management Journal,* the *Strategic Management Journal,* the *Journal of Business Venturing,* and the *Journal of Environmental Economics and Management.*

CHAPTER 1

INTRODUCTION TO SUSTAINABLE VENTURING

INTRODUCTION

In 1977, Colorado entrepreneur Steve Demos founded WhiteWave soy foods with the dream of bringing the nutritional and environmental benefits of soy food products to U.S. consumers. Starting with $500 in capital, Demos began making tofu products in his bathtub and walked them to local customers in a red wagon. By the mid-1980s, White Wave was shipping soy products nationally to over 1000 natural food stores, but struggled financially. White Wave was still a small company with limited appeal to larger markets. Soymilk, one of its primary product lines, remained hidden on the dry shelves in unappealing aseptic boxes. That's when Demos decided to combine his passion for soy products with professional marketing and entrepreneurial skills. White Wave reformulated its soymilk and created the Silk brand. The product was eventually placed beside milk in the refrigerated aisle. With the new product, appealing branding, and attractive shelf space, sales grew rapidly and expanded beyond natural food stores to national grocers. Dean Foods, the largest producer of milk in the United States, purchased the company at a valuation of over $200 million. Today, the White Wave Division leads Dean Foods' forays into natural and organic food products, holding brands such as Silk, Horizon Organic, and Land O'Lakes.[1, 2] Demos, meanwhile, continues his entrepreneurial endeavors in natural foods, having recently founded NextFoods. The company's first product, GoodBelly, is a popular probiotic fruit drink that builds digestive health.

After spending his college summers selling books door-to-door, a young Quayle Hodek turned his sights on renewable energy. Founding Renewable

Choice in 2001, Quayle applied his sales abilities to the marketing of renewable energy and hired college students to sell renewable energy credits to residential customers during their summer break. No easy task, but the strategy was successful enough to keep the company going. Then interest and demand from commercial customers and green builders bloomed. Within the span of only a few years, Renewable Choice had helped industry leaders such as Whole Foods, Vail Resorts, and Gander Mountain offset their electric power usage with renewable energy and carbon offsets. The company now employs more than twenty-five people and has been a driving force in the marketing of renewable energy.

Responding to the lack of reliable electrical power infrastructure in Kenya, entrepreneur Simon Mwacharo began experimenting with wind turbines. He enlisted friends with electrical experience to help design charge controllers, generators, fiberglass blades, and towers. He created CraftSkills East Africa Limited, which builds and installs low-cost, locally designed wind turbines. The company markets to rural homes, schools, hotels, health facilities, and shops. They have installed over eighty wind turbines, bringing power, lighting, security, and cell phone charging to customers. In one case, they built a wind turbine and distribution system that provides power to a thirty-hut village in a remote valley of Kenya. To add value, the company connects its devices to water pumps, LED lighting systems, and battery-charging centers.[3, 4]

These entrepreneurs and the companies they created are not just individual stories of success, they portend the next entrepreneurial revolution—that of sustainable venturing. Driven by fundamental trends in our global economy, sustainable entrepreneurs are creating a more ecologically, socially, and economically sustainable way of life. They do so, not so much at the expense of profit, but at the behest of it. For, as resource scarcity, population, and climate change alter the economic fabric of global society, new opportunities for profit are arising. The sustainable entrepreneurs who recognize these changes and create new technologies, products, and business models that enhance the sustainability of our global resources stand to benefit. Entrepreneurs helped create our present, and sustainable entrepreneurs will pioneer our future.

ENTREPRENEURS: OUR MODERN-DAY PIONEERS

Entrepreneurship, *the process of creating value by bringing together a unique combination of resources to develop and pursue an economic opportunity,* is one of the most powerful forces in the world today. From Ford Motor Company, to Apple Computer, to Google, entrepreneurial companies have driven technological development, changed the way we live our lives, and fundamentally altered the course of human history. From food production, health care, and transportation, to modern conveniences and entertainment, entrepreneurs in many industries have created better products and services and delivered them in increasingly efficient ways. Thanks to entrepreneurs and the market systems that drive them, developed societies live today in a world of prosperity and capability only dreamed of by earlier generations.

Entrepreneurs are powerful drivers of positive change because they *tend to be rewarded for creating value for society and its members.* This was the fundamental realization of famed economist Adam Smith when he stated that "by directing that industry in such a manner as its produce may be of greatest value, he intends only his own gain and, he is in this, as in many other cases, led by an invisible hand to promote an end which was no part of his intention. . . . By pursuing his own interest he frequently promotes that of society more effectually than when he really intends to promote it."[5] In other words, when entrepreneurs create value by offering products and services that people want and need, they increase their chances of realizing economic profits while also making society better off.

The potential to achieve profit is commonly referred to as *economic opportunity.* At any given time, opportunities exist within the system of rewards present in the global economy. Opportunities can also be created through changes in the system of rewards over time. Many opportunities are hidden or otherwise difficult to perceive or pursue, but that is what allows their continued existence and creates the potential for profit. The opportunities that are effortlessly observed and easily exploited tend to disappear quickly due to the competitive nature of markets, so they are transient in the sense that they can both disappear quickly and be created as the result of fundamental changes in demand, supply, and other conditions. Such opportunities are windows that open and close according to both the rewards available at any given time and the extent of competition for those rewards.

Opportunities have also been referred to as *market gaps or imperfections,* implying the potential to fulfill new customer demand conditions or existing demands in new ways. As such, economic opportunities are fundamentally linked to the creation of value for customers, or society, more broadly. Opportunities are, in their essence, problems waiting to be solved, holding rewards for those who solve them. The sustainability of our environmental resources is one of the biggest problems, and opportunities, facing society today.

"HOUSTON, WE HAVE A PROBLEM"

In 1968, when lunar astronauts snapped the first photograph of Earth rising from the moon, one thing became immediately clear: We all live on a single planet in an otherwise rather desolate expanse of space.[6] From the ground, Earth appears vast and unending. From space, it appears small, isolated, and precious. The fact is that we have the resources of only one Earth. With the exception of the input of solar energy, our planet is what systems-thinkers call a *closed system:* It has limits, and it is constrained. Earth's resources are scarce, and they are valuable because they provide the means by which we eat, drink, walk, play, laugh, and love.

Ecosystem services are the services that Earth's ecological systems provide to human existence. In other words, they represent the value that the ecosystems create for human society. Ecosystem services include a host of benefits provided by the natural environment, including regulation of the air we breathe, the stabilization of our climate, the supply of water, the production of food and raw materials, and recreation. One estimate of the value of these services places their annual value at $33 trillion dollars, greater than the global gross national product.[7] Regardless of whether such numbers are accurate, our economy and civilization are fundamentally reliant on ecosystem services and on our planet's ecological resources.

Yet the *ecological footprint* of our economic activities is arguably greater than the capacity of the ecological systems that provide those services.[8] Current estimates of humanity's ecological footprint, the burden of our activities on the environment, suggest that it would require 1.4 Earths to sustain the current level of economic activity.[9] While climate change seems to present the most ominous challenge, the loss of biodiversity, diminishing freshwater supply, soil erosion, air pollution, infectious disease, and degradation of fisheries remain serious and costly issues. Overuse of natural systems foretells the eventual diminishment of the valuable environmental resources (often referred to as *natural capital*) upon which we rely. The number of people that natural capital can support, and their quality of life, depends on the magnitude of the earth's resources and the way in which we use them. Many argue that the global economy, as currently constructed, is not ecologically sustainable—that is, it does not stand up to the test of *sustainable development,* which is to meet the needs of the present without compromising the ability of future generations to meet their own needs.[10] Embedded in the idea of sustainable development is the realization that any economic activity that is not socially and ecologically sustainable will not be economically sustainable in the long run either. Economic development, social value creation, and environmental preservation are linked at their core.

THE BIGGEST OPPORTUNITY

Sustainable venturing is the process of creating socially and ecologically sustainable value by bringing together a unique combination of resources to pursue an economic opportunity. *Ecologically sustainable value* is value created for society and its members that simultaneously sustains or enhances the social and ecological resources of the planet. Sustainable entrepreneurship can be seen as a subset of the larger concept of entrepreneurship, but as society adapts to the realities of the present and future, the two will become increasingly aligned. In this book, we use the terms *sustainable venturing, sustainable entrepreneurship,* and *environmental entrepreneurship* as similar or equivalent. We also discuss the concept of social entrepreneurship, as we see social issues as an important aspect of sustainable development.

If society is to be successful in the long term, the system of rewards must create a context in which unsustainable entrepreneurship is not economically profitable. More important, as the system of rewards that motivates entrepreneurs evolves through increasing recognition of our global challenges, opportunities for sustainable entrepreneurship will grow substantially. This process of becoming more environmentally aware, and open to new economic opportunities as a result, is already underway.

In 1990, Francis Cairncross of the *Economist* wrote that the environment may turn out to be "the biggest opportunity for enterprise and invention the industrial world has ever seen."[11] Today, Ms. Cairncross's prognostication is manifest in a multitude of industries around the world. Innovative entrepreneurs across the globe have started businesses that are creating rewards for them and for environment. These sustainable entrepreneurs are not easily categorized into a single type, nor do they fit within any single industry. Rather, they come in many shapes and sizes, and span a host of sectors. They do share the common attribute of offering new products, services, or business models that increase our social and ecological sustainability.

Most celebrated and exciting of these sustainable ventures are the technology start-ups with high growth potential, most commonly referred to as clean-tech (or green-tech) ventures. Some clean-tech ventures hold the prospect of fundamentally changing the delivery of products and services and have been a magnet for recent investors. Tesla Motors in California offers a completely electric car that can accelerate from 0 to 60 miles per hour in 3.9 seconds while generating one-tenth of the pollution of comparable sports cars. Numerous companies are developing advanced lithium batteries with higher power and longer life, potentially enabling the production of lower-cost electric vehicles while also reducing waste and materials use. Clipper Windpower developed a wind turbine technology that reduces loading on its internal mechanism, thereby increasing reliability. The result is lower turbine operating costs and more competitive prices for wind-generated electricity. Clean-energy ventures have garnered the largest share of clean-tech investments, but start-ups in other sectors have been robust as well. The clean-tech revolution comprises companies in energy efficiency, energy storage, water purification and management, advanced materials, recycling, agriculture, and manufacturing.

But sustainable entrepreneurs don't just include high-growth technology ventures. A host of others are founding small businesses and lifestyle companies that provide a living while allowing them to pursue their passions. Nate Burger, the Eco-Handyman, provides a variety of household services with a green flair. Recent trends toward energy-efficient and nontoxic renovations have allowed him to build a healthy contracting business. Entrepreneur and innovative manager Blake Jones founded Namaste Solar after spending years installing remote solar systems in Nepal. Namaste Solar has become renown for its collaborative, equitable, and fun company culture, as well as a leading installer of solar energy systems in homes and commercial buildings.

Other sustainable ventures aren't driven by complex technology but have become significant and notable businesses. Horizon Organic grew quickly after introducing its line of organic dairy products with the attractive "happy cow" icon. Horizon is now the leading brand of organic milk in the United States and helped convert hundreds of thousands of acres to organic farming. Evolution Markets, an emissions-allowance trading firm, employs over eighty energy and emissions brokers worldwide, has built a reputation as one of the top emissions-trading and financial companies in the world, and has experienced substantial growth as a result of global climate change policy. Rocky Mountain Sustainable Enterprises has built a substantial business by recycling used cooking oil into biofuel. Their RecycOil service collects used cooking oil from restaurants, processes the waste into diesel, and offsets the consumption of fossil fuels.[12]

Some sustainable entrepreneurs are of the corporate type. Kevin Beaty and his team at Eaton Company helped develop a transmission system for diesel electric hybrids that is now utilized by FedEx and the United Parcel Service. Starting about five years behind their competition, Eaton Company succeeded by letting customer input drive technology development. Their hybrid systems can reduce fuel consumption in stop-and-go truck applications by as much 60 percent. Toyota's and Honda's hybrid automobile offerings have had similar success, offering substantial increases in fuel economy. But probably the biggest corporate effort into environmental entrepreneurship is that of General Electric (GE). In 2005, GE surprised the world when it announced its ecomagination initiative. Chief Executive Officer Jeff Immelt announced that "green is green" and that producing environmentally friendly and energy-efficient products would drive revenue growth and profitability in their company. The ecomagination initiative launched technological and product development in a multitude of sectors including compact fluorescent bulbs, energy star appliances, locomotive engines, solar modules, and energy-efficient home improvement loans.

In other cases, the products sustainable entrepreneurs create are not as obviously sustainable, but their delivery or production are. New Belgium Brewing brews its high-quality beers at a highly efficient and 100 percent wind-powered facility. The company uses anaerobic digesters to transform its waste stream into energy for the brewery. It also uses 20 percent less water in the brewing process and has one of the most energy-efficient brewing facilities in the world. The company has also developed a values-driven culture of ecological sustainability and community involvement which helps drive innovation and reduce ecological impact. Outdoor apparel manufacturer Patagonia takes a similar approach. "Committed to the Core," the company, is an example of what environmental entrepreneurs can accomplish. From their efforts to preserve important ecosystems to the promotion of organic cotton, Patagonia has been a leader in implementing and promoting more sustainable business practices.

DRIVERS OF OPPORTUNITY: THE SIX C'S

The multitude of opportunities for sustainable venturing is being driven by fundamental trends in our global society and economy. Though these trends are complex and multifaceted, it is useful to think of them as the six C's: Costs, Capital, Consumers, Climate, Consciousness, and Convergence.[13]

TABLE 1.1 Drivers of Opportunity—The Six C's

Driver	*Description*	*Examples*
Costs	Declining costs of clean technology	Declining costs of wind power
	Increasing costs of natural resources	Increasing costs of oil
	Increasing internalization of externalities	Increasing costs of carbon emissions
Capital	Increasing investor and government funding of social and environmental enterprises	Clean-tech venture capital Social capital and impact investing Government stimulus packages aimed at green technology
Consumers	Increasing desire to purchase green products	Organic foods, fair-trade coffee, renewable energy
Climate	Global climate change	Carbon taxes and cap-and-trade regimes for greenhouse gas emissions
Consciousness	Increasing public awareness of environmental and social issues	Changing attitudes and beliefs about environmental effects and how those effects impact economic activity
Convergence	Increasing alignment between economic incentives and positive social and environmental outcomes	Increased regulation, market incentives, and stricter property rights for environmental resources such as fisheries

COSTS

The foremost trends in the green revolution can be seen in changing costs, and there are three important cost drivers for sustainable venturing in the world today: declining clean-technology costs, increasing natural resource costs, and increased internalization of environmental costs. First and perhaps most important, the cost of clean technology is falling rapidly. This is probably nowhere more evident than in the clean-energy sector. The cost of generating utility-scale wind power, for example, is now quite competitive with the production of electricity through the burning of coal. By building large farms of wind turbines, entrepreneurs have been able to substantially lower the costs of generating wind power. Second, the cost of resources is trending upward.[14] Though energy prices rise and fall with the global economy, many predict that the costs of fossil fuels will rise substantially over the next few decades. Increasing fossil fuel prices will make clean energy even more attractive, and will drive entrepreneurs to build more and more businesses that deliver products and services with minimal energy and resource use. Third, emerging national and international policies will increase the costs of doing business for those that harm the environment. Economists refer to this as *internalizing externalities,* which means that businesses that damage the environment will increasingly bear the burden of their actions in their bottom line. Together, these cost drivers are changing the competitive rules of the game, making sustainable entrepreneurship ever more attractive as clean technologies become more efficient, resource prices rise, and governments require producers to bear the full costs of their production.

CAPITAL

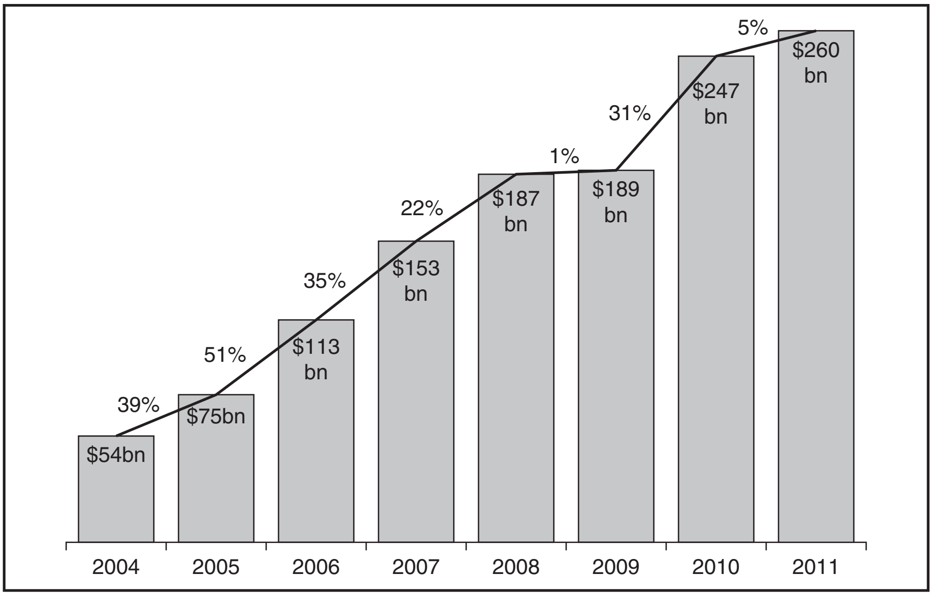
Private investors have determined that there is money to be made through sustainable venturing. Some investors who may see nothing but dollars are quite happy to participate in endeavors that help preserve our environment. Others are more driven by the prospect of social or ecological sustainability, but still care a great deal about their investments and returns. Private investors putting dollars into environmental ventures include venture capitalists, private equity firms, growth funds, corporations, angel investors, and a myriad of self-funded entrepreneurs. Venture capitalists invest in high-potential firms, typically those with new technologies. In the past decade, venture capital investment in the global clean technology sector rose from almost nothing to $8.9 billion in 2011.[15] Leading venture capital firms such as Khosla Ventures and Kleiner, Perkins Caufield and Byers (KPCB) have raised hundreds of million of dollars for specialized green investment funds and have been rapidly investing in emerging clean-tech companies. Expansion funds take these ventures to the next level, funding firms with proven technologies or products in need of growth financing. Individual angel investors typically fund environmental entrepreneurs in earlier stage companies with smaller investments. Large corporations such as Walmart, GE, and Sun Microsystems have been investing in both internal and external entrepreneurs with ideas for increased efficiency and enhanced sustainability. In total, global investment in clean energy was estimated to be $260 billion in 2011 (see Figure 1.1).

Governments are also getting into the act. National, state, and local governments have initiated a host of tax incentives, rebates, regulatory requirements, research programs, and even direct investments that financially support environmental entrepreneurs. Their motivations for such investments vary, but most programs are directed at promoting clean industries, creating employment opportunities, advancing energy independence and national security, or helping solve the climate change issue. Germany, for example, has led the world in support of rooftop solar electric systems through tariffs that return money to owners of the systems. By offering this support early, the German government has expedited the creation of a number of new companies that are leading the way in solar technology. State-level renewable portfolio standards in the United States require electric utilities to produce a certain percentage of their power from renewable sources. Such states have witnessed the proliferation of start-ups in the renewable energy market.[16]

CONSUMERS

Consumers have also been a driving force in the environmental entrepreneurship revolution. Across the world, consumers are demanding more accountability on

FIGURE 1.1 Total New Investment in Clean Energy, 2004—2011



Source: Bloomberg New Energy Finance: Green Investing 2011, Global Trends in Clean Energy Investment, 12 January 2012. Bloomberg, LP.

the part of producers, driving fundamental changes in behavior. *Negative screening* occurs when consumers avoid certain products or brands because of their environmental impact. Purchasing fair-trade coffee, renewable energy, or dolphin-safe tuna are prime examples of consumer purchasing decisions based on environmental factors. Some consumers are willing to pay more for environmentally differentiated products, potentially providing extra margins to the entrepreneurs who produce them. These consumers "vote with their wallet" to support their core beliefs and values. Companies like Renewable Choice, Green Mountain Coffee, and Terrapass have all benefited from these changing consumer behaviors. Other customers don't necessarily care about environmental impacts of products, but look for a host other benefits associated with these products. For example, the green building market has expanded rapidly, not so much because customers want a green building per se, but rather because they desire increased energy efficiency, lower toxicity, enhanced comfort, and greater workforce productivity. The same may be said for organic foods, as most customers purchase organic foods first and foremost because they are concerned about their, and their children's, health.

CLIMATE

Climate change is the most prevalent environmental issue in the world today. The idea that human burning of fossil fuels (coal, oil, and natural gas) could produce enough greenhouse gases to alter the climate was introduced decades ago. Scientists estimate with increasing certainty that the global climate is changing, that the change is caused by the human production of greenhouse gases, and that it will have substantial effects on our society and economy. While many wrestle over the science of climate change or agonize about its negative impacts, environmental entrepreneurs see an opportunity to use business as a vehicle for making positive environmental impact. Regardless of the arguments over the science, we are already in a system of increasing greenhouse gas management and control—what many call a "carbon-constrained world." What this means for business is an increasing "cost of carbon" wherein companies, and even individuals, pay for their greenhouse gas emissions. Such costs are changing the fundamental incentives in the marketplace so that clean, low-carbon technologies and products are more poised to win the competitive game. Innovative entrepreneurs recognize the value of low-carbon technologies and may sell their carbon credits on global and national carbon markets.

CONSCIOUSNESS

Fundamental changes in global consciousness with respect to environmental issues are driving sustainable venturing. At the foundation of this new consciousness is increasing scientific and popular knowledge about issues such as climate change, population growth, and ecological degradation. But this consciousness goes beyond just knowledge to changing values about what is important, appropriate, and rewarded in the economic system. While individual action is driven by the present system of rewards, future action is a function of the changing beliefs held by society, and those beliefs eventually impact the rules of the game in which entrepreneurs participate. The process of changing beliefs and attitudes and adjusting rules to match those beliefs is slow, but continuous. Beliefs tend to change over decades and sometime centuries, but they can also change rapidly, typically in response to a major event or crisis. Today, the tension between old ways of thinking about industry and the economy and emerging beliefs about the impact of our economy on the environment is evident to those with the awareness to see it. How rapidly the tension will break remains to be seen, but the convergence between new beliefs and the reality of our present ecological condition is already underway.

CONVERGENCE

Convergence is the increasing alignment between the institutions (or rules of the game) that determine competitive behavior and the modern realities of our planet. In other words, convergence is the increasing alignment between the nature of opportunity in our economic system and the creation of socially and ecologically sustainable value. As global consciousness of ecological degradation increases, the incentives for more sustainable action will grow, and capitalism will become increasingly aligned with good social and environmental outcomes. Such convergence is the locus of opportunity for entrepreneurial action, and the realm of the sustainable entrepreneur. Whether it is global climate change policy, tax credits for renewable energy, rising energy prices, or increasing recognition of the lower costs of sustainable technologies, entrepreneurs in the post-industrial world will be increasingly rewarded for their sustainable actions.

But the best sustainable entrepreneurs don't just wait for the convergence of economics incentives and ecological sustainability. Rather, they initiate the changes that help reward them for their more sustainable business models. They take it upon themselves to change the rules of the game in their favor and in favor of the environment. In the face of entrenched interests, this can be a difficult and time-consuming process, but the wind of sustainability and the societal value created by more sustainable business models can help entrepreneurs change the system of rewards. These entrepreneurs also find allies in the myriad of government agencies and nonprofits that share their visions and goals. Perhaps most important, their passion, belief, and action are critical elements in overcoming the status quo.

MYTHS ABOUT SUSTAINABLE VENTURING

The popularity of the concept of sustainable entrepreneurship is rapidly increasing. Indeed, many are excited about the prospect entrepreneurship holds for the environment. But many also misconceive what it is and how the process might work to help build a more sustainable economy. The following are common myths about sustainable entrepreneurship.

Myth 1: Sustainable venturing is about saving the world, not making a profit.

Doing the right thing and contributing to sustainability is the objective of many sustainable entrepreneurs, and their passion for implementing change can be a strong driving force for entrepreneurs, their employees, and their investors. But business activity that is not profitable is not economically sustainable, nor is it particularly replicable. There is probably no better motivation for entrepreneurs than to prove that you can make money while contributing to sustainability.

Myth 2: You have to want to save the world to be a sustainable entrepreneur.

Contrary to popular belief, not all sustainable entrepreneurs are passionate about resolving our environmental challenges. Some just see the business opportunities present in the transition to a sustainable economy and act more in response to profit signals than to a personal desire for change. If they are successful, the outcome can be similar, regardless of the initial motivation.

Myth 3: If I just do well for the environment, I'll make money.

Green is not always green. The reality of our current economic system is that many actions that would be helpful to the environment just aren't attractive from a business perspective, at least today. Some business ideas that are better for the environment will not yet be successful because they are not yet attractive economic opportunities. The key is to find where more sustainable business models intersect with the potential for profitability or to change the rules of the game to make it so.

Myth 4: Consumers will prefer my green product over less green options.

A percentage of consumers buy products because of their environmental values and are even willing to pay more. This represents a real green opportunity, although a niche one. Most customers care much more about product performance and quality. Sustainable entrepreneurs who wish to access broader markets need to provide and communicate a value proposition that goes beyond green. Fortunately, value drivers such as energy efficiency, healthfulness, and lower operating costs tend to be associated with green products. Selling these value attributes is usually the best way to market a green product or service.

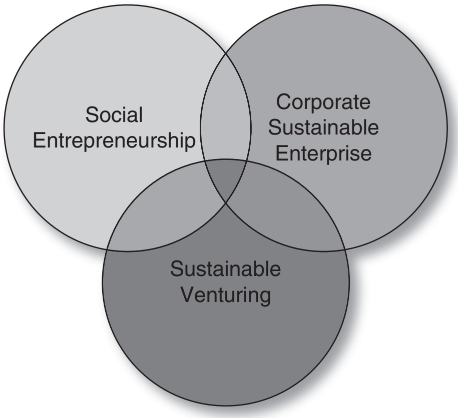
Myth 5: Once I find the right environmental opportunity, the rest is easy.

The world will not beat a path to your door just because you build a better mousetrap. Sustainable entrepreneurship, like most entrepreneurial endeavors, requires good business strategy and effective implementation. You'll want a source of competitive advantage, like a good brand name, differentiated product, or intellectual property. The best sustainable entrepreneurs also attract teams of intelligent and experienced professionals who complement their skills. Starting a business is no easy matter, and sustainable entrepreneurs need to be prepared for the challenge.

THE DOMAINS OF SUSTAINABLE ENTERPRISE

As you learn more about sustainability and business, you will come across a variety of terms, and it is useful to understand the differences and similarities between them. In this process, it is important to recognize that many of the terms are just now emerging, and definitions are not always clear. In addition, the scope of issues and methodologies associated with sustainability in business is large, so there is a great deal of ground to cover. One way to think about the various terms is shown in Figure 1.2. The figure shows the overlapping domains of sustainable enterprise— sustainable venturing, social entrepreneurship, and corporate sustainable enterprise—illustrating how the concepts are related. *Corporate sustainable enterprise* tends to focus on the challenges and techniques of managing sustainability issues in large companies. Managing risks, measuring emissions and effects, and generating policies and systems to accommodate sustainability are foremost in the field of corporate sustainable enterprise. Both social entrepreneurship and sustainable venturing tend to focus more (though not exclusively) on innovative new organizations that solve social and environmental challenges. The field of *social entrepreneurship* emerged at least partially from nonprofit organizations, so there is a tendency to focus less on profit motivations and more on impact. In contrast, *sustainable venturing* (or entrepreneurship) is often more closely affiliated

FIGURE 1.2 Domains of Sustainable Enterprise



with market-driven enterprises and the increasing alignment between economic motivations and social and environmentally sustainable outcomes.

In addition, the term *social entrepreneurship* is popularly affiliated with enterprises that create (or desire to create) social value, whereas the term *sustainable entrepreneurship* is often associated with enterprises that reduce environmental degradation or improve ecological systems. Our perspective in this text is that the term *sustainability* encompasses social, environmental, and economic aspects. Perhaps most important, we believe that social and environmental issues are so intertwined that it is impossible to achieve sustainable development without considering both. Social problems can exacerbate environmental challenges as the poor struggle to survive and social systems fail to recognize the value of critical environmental resources. Conversely, environmental problems can have serious effects on social conditions, as natural resources necessary to support human populations degrade. The destruction of forests, for example, can cause desertification and flooding, as well as diminish food resources for local populations. Global climate change can have similar effects and can cause decreases in agricultural productivity. Such changes can result in armed conflict over resources and migrations that exacerbate social challenges. Furthermore, it is difficult to conceive of economically sustainable societies in which social or environmental problems are pervasive.

FOCUS OF THE BOOK

This book focuses on market-driven business ventures, which capture opportunities emerging from environmental challenges. We embrace the concept of social entrepreneurship as part of the domain of sustainable venturing, particularly for its unique focus on mission-driven organizations that have the potential to enhance both social and environmental sustainability. We also see the methods of corporate sustainable enterprise as valuable tools in the process of sustainable venturing. In building a sustainable venture, the methods of large corporations in addressing sustainability can be useful to the budding sustainable entrepreneur. Perhaps most important, we emphasize economic opportunities that are arising from the increasing challenges of a planet with limited resources. We see the sustainable entrepreneur and venture as responding to changing economic dynamics, using market-driven action as a means to resolve these challenges.

The next chapter provides an economic systems view of environmental challenges as a foundation for understanding the nature of opportunities for sustainable venturing. Chapter 3 translates some of these system-level perspectives into specific strategies that the sustainable entrepreneur can use to overcome barriers to the effective functioning of markets and to enable the sustainable venture. Chapter 4 focuses on the concept of social entrepreneurship, explaining the potential for entrepreneurs to solve social problems through mission-driven endeavors. Chapter 5 discusses the field of environmental marketing, addressing five opportunities to market products and services based on their environmental or social attributes. Chapter 6 then discusses the methods and considerations necessary to start and build the sustainable venture. Chapter 7 provides an overview of venture finance for the sustainable entrepreneur.

Summary

As our global society struggles with the challenges of environmental degradation and ecological sustainability, we increasingly look toward the process of entrepreneurship as the answer. Sustainable venturing holds the potential for innovative solutions to our most pressing challenges. The revolution of sustainable entrepreneurship is already underway, and there is much more to come as society adjusts to the reality of post-industrialism. The potential convergence between the creation of value for society in the realm of the environment and economic returns for entrepreneurs is what makes sustainable entrepreneurship one of the most exciting frontiers in today's world. Properly incentivized by the economic system, sustainable entrepreneurs can have their cake and eat it too. They can be passionate about resolving our environmental challenges and put profits in their pockets for their efforts.

Endnotes

1. Lea Goldman and Tom Post, "A Cry in the Wilderness," *Forbes* 165, no. 11(2000): 322–324.
2. G. Pascal Zachary, "Why the Soy Milk King Still Reigns," *Business 2.0* 5, no. 3 (April 2004): 73–74.
3. Craftskills, East Africa Limited, at http://craftskillseastafrica.com/b-one-default.html, accessed January 20, 2012.
4. J. Schmitt, "Craft Skills East Africa Limited," in T. J. Dean and M. D. M. Birnbaum, *Cases with a Conscience* (Boulder, CO: Sustainable Venturing Press, 2011.

5. Adam Smith, *The Wealth of Nations* (London: Strahan and Cadell, 1776), p. 456.
6. Steve Connor, "Forty Years Since the First Picture of Earth from Space," *The Independent*, January 10, 2009, available online at http://www.independent.co.uk/news/science/forty-years-since-the-first-picture-of-earth-from-space-1297569.html.
7. R. Costanza, R. d'Arge, R. de Groot, S, Farber, M. Grasso, B. Hannon, K. Limburg, S. Naeem, R. V. O'Neill, and J. Paruelo, "The Value of the World's Ecosystem Services and Natural Capital," *Nature* 387 (May 1997): 253–260.
8. M. Wackernagel and W. Rees, *Our Ecological Footprint: Reducing Human Impact on the Earth* (Gabriola Island, BC: New Society Publishers, 1996).
9. Redefining Progress website, http://rprogress.org/ecological_footprint/about_ecological_footprint.htm, accessed January 8, 2013.
10. G. Bruntland, ed. *Our common future: The World Commission on Environment and Development* (Oxford, Oxford University Press, 1987).
11. Francis Cairncross, *Costing the Earth: The Challenge for Governments, the Opportunities for Business* (Boston: Harvard Business School Press, 1992), p. 177.
12. "Used Cooking Oil Recycling," http://www.recycoil.com/ucor.html, accessed January 20, 2012.
13. Based on R. Pernick and C. Wilder, *The Clean Tech Revolution* (New York: Collins Business, 2008).
14. R. Dobbs, J. Oppenheim, and F. Thompson, "Mobilizing for a Resource Revolution," *McKinsey Quarterly* (January 2012): 1–15. Available online at https://www.mckinseyquarterly.com/Mobilizing_for_a_resource_revolution_2908.
15. Whitney Bennett Michael, "Corporate Cleantech M&A Grew 153 Percent in 2011 Totaling $41.2 Billion," press release, January 5, 2012, The Cleantech Group.
16. D. Yergin, *The Quest: Energy, Security, and the Remaking of the Modern World* (New York: Penguin, 2011).